

Implementation Plan: AI-Enabled Detection of Exoplanets from Noisy Astronomical Light Curves

1. Executive Synopsis and Mission Parameters

The Bharatiya Antariksh Hackathon 2026, initiated by the Indian Space Research Organisation (ISRO) in collaboration with Hack2skill, serves as a premier crucible for advancing national capabilities in space technology, planetary science, and artificial intelligence.¹ Within this national-level innovation challenge, Problem Statement 7 defines a critical mandate: the conceptualization, development, and deployment of an Artificial Intelligence (AI) driven data analysis pipeline capable of automatically detecting exoplanet transit signals from highly noisy astronomical light curve data.³ This document constitutes the exhaustive, technically rigorous execution plan required to master this challenge, meticulously designed to operate within the specific computational constraints of a local workstation powered by an NVIDIA RTX 4080 Graphics Processing Unit (GPU) and a latest-generation Intel Core i9 processor.⁴

The primary objective of this initiative extends beyond mere binary detection. The mandated AI pipeline must achieve a highly nuanced classification framework capable of differentiating true planetary transits from a multitude of astrophysical false positives and instrumental anomalies present in crowded stellar fields. Specifically, the system must categorize periodic luminosity dips into four distinct classes: genuine exoplanetary transits, eclipsing binary stellar companions, stellar blends (arising from foreground or background sources contaminating the aperture), and other astrophysical phenomena such as rotational modulation caused by starspots.

Furthermore, the hackathon's expected outcomes demand a robust transition from classification to quantitative astrophysical characterization. The developed algorithm must accurately estimate critical transit parameters—specifically the orbital period, the transit duration, and the transit depth—while simultaneously providing rigorous signal-to-noise ratios (SNR), statistical significance levels, and confidence intervals for all identified events. The culminating deliverable must include sophisticated visualizations of the phase-folded light curves alongside the detected signals, accompanied by a precise, maximum three-page scientific report detailing the methodology, the computational assumptions, the software libraries utilized, and the rigorous mathematical frameworks employed to estimate uncertainties.

To achieve this multifaceted objective within the rigid 30-hour developmental window of the Grand Finale scheduled for August 6-7, 2026, the architectural blueprint must intertwine cutting-edge astrophysics with highly optimized deep learning paradigms.⁶ The ensuing sections delineate the theoretical foundations, the comprehensive data acquisition strategy encompassing both the primary Transiting Exoplanet Survey Satellite (TESS) archives and supplementary datasets, the highly parallelized preprocessing pipelines, the hybrid neural

network topologies, and the extreme hardware optimization protocols required to dominate this challenge.

2. Astrophysical Theory and Transit Photometry Dynamics

The detection of exoplanetary bodies from photometric time-series data relies on the transit method, a mechanism wherein an orbiting planet crosses the stellar disk of its host star along the observer's line of sight, thereby obscuring a minute fraction of the emitted stellar flux.⁸ This partial occultation results in a characteristic, periodic U-shaped dip in the recorded light curve. The extraction of this signal from raw, uncalibrated data forms the crux of Problem Statement 7.

The Mathematics of the Transit Signal

The fundamental physics governing the transit phenomenon dictates that the measured exoplanet transit depth (δ)—the maximum relative decrease in stellar flux during the event—is directly proportional to the square of the ratio of the planetary radius (R_p) to the stellar radius ($R_\star$).¹⁰

$$\delta \approx \left(\frac{R_p}{R_\star} \right)^2$$

This geometric approximation assumes a uniform stellar disk. In reality, stars exhibit limb darkening, where the center of the stellar disk appears brighter than the edges due to the optical depth of the stellar photosphere. Consequently, the true transit light curve is not a perfect box-car shape but exhibits a curved bottom and sloped ingress and egress phases. The analytical modeling of these light curves requires the implementation of limb-darkened transit models, often utilizing quadratic or non-linear limb darkening laws to accurately fit the transit shape and derive precise parameters.

The transit duration (T_{14}), defined as the total time from first contact (ingress) to fourth contact (egress), is a function of the orbital period (P), the semi-major axis (a), the stellar radius ($R_\star$), the planetary radius (R_p), the orbital inclination (i), and the orbital eccentricity (e). For a circular orbit ($e = 0$), the duration is approximated by:

$$T_{14} \approx \frac{P}{\pi} \arcsin \left(\frac{\sqrt{(R_\star + R_p)^2 - (a \cos i)^2}}{a} \right)$$

These mathematical relationships form the bedrock of the parameter estimation requirement

mandated by the problem statement. The AI pipeline must not only detect the event but also interface with physical models to extract P , T_{14} , and δ with mathematically quantified uncertainties.

Signal Degradation and Astrophysical Noise Profiles

The observational reality of transit photometry is fraught with severe signal degradation. The photometric data captured by the TESS charge-coupled devices (CCDs) are subjected to continuous noise from multiple sources, necessitating advanced filtering techniques before AI ingestion.¹⁰

Noise Categorization	Physical Origin and Characteristics	Impact on Photometric Time-Series
Instrumental Noise (False Alarms)	Spacecraft momentum dumps, thermal fluctuations, thermal jitter, and scattered light from the Earth and Moon entering the telescope aperture.	Produces abrupt, non-physical spikes or severe baseline shifts in the flux that can trigger false transit detections if not properly masked or detrended.
Photon Shot Noise	Inherent statistical fluctuations in the arrival rate of photons from the target star, governed by Poisson statistics.	Imposes a fundamental floor on the achievable photometric precision, particularly degrading the signal-to-noise ratio for faint stars or exceptionally small terrestrial planets.
Intrinsic Stellar Variability	Rotational modulation of active stellar regions (starspots), stellar pulsations, and localized magnetic flares.	Creates long-term, quasi-sinusoidal variations in the baseline flux that can completely mask the minute, high-frequency signals of transiting exoplanets.
Stellar Blending (Contamination)	Unresolved foreground or background stars falling within the same pixel aperture as the target star	Dilutes the true transit depth, causing Jupiter-sized planets to masquerade as Earth-sized

	due to crowded stellar fields.	planets, or causing background eclipsing binaries to mimic planetary transits.
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The Taxonomy of Periodic Signals

To satisfy the explicit requirement of developing a classification framework to categorize light curve dips, the AI architecture must be trained to recognize the subtle morphological differences between genuine planets and astrophysical impostors.

- Planetary Transits:** Characterized by a flat-bottomed or slightly curved U-shape (depending on limb darkening), strict periodicity, and a depth typically less than 1-2%. Planetary transits do not exhibit secondary eclipses (unless the planet is exceptionally hot and luminous), and the depth remains constant across different wavelength bands (achromaticity).
- Eclipsing Binaries (EBs):** Binary star systems where the orbital plane is aligned with the observer. These typically exhibit deep, sharp V-shaped primary eclipses and distinct secondary eclipses occurring at half-phase (for circular orbits). A hallmark of EBs is the odd-even effect, where alternating eclipses possess different depths due to the differing luminosities of the primary and secondary stars.¹⁰
- Stellar Blends (Background Eclipsing Binaries):** A faint, spatially unresolved eclipsing binary located behind the target star. The bright target star dilutes the deep eclipse of the background binary, resulting in a shallow dip that closely mimics a planetary transit. Detecting these requires analyzing the centroid shift of the target during the transit event, as the center of light will move slightly toward the bright target and away from the faint background binary during the eclipse.
- Other Astrophysical Categories (Starspots):** Large, cool regions on the stellar surface rotating in and out of view. These produce periodic, highly variable sinusoidal dips that lack the sharp ingress/egress boundaries characteristic of transiting spherical bodies.

The periodicity matches the stellar rotation period rather than a Keplerian orbital period. The deep learning model must be optimized to map these complex, high-dimensional temporal features into the four defined categorical spaces, demanding a sophisticated interplay of convolutional feature extraction and sequence modeling.

3. Comprehensive Data Acquisition and Curation Strategy

The foundation of the proposed AI system is the robust curation, ingestion, and cross-validation of high-fidelity photometric data. The problem statement explicitly directs the utilization of the Transiting Exoplanet Survey Satellite (TESS) raw light curves accessible via the publicly available repository at the Space Telescope Science Institute (STScI).¹³ However, to build a truly robust, generalizable AI model, the data strategy must transcend a single catalog

and incorporate a diverse array of supplementary datasets.

Primary Dataset: The TESS Input Catalog and Candidate Target List

The primary resource for target selection and foundational stellar parameters is the TESS Input Catalog (TIC), a comprehensive database compiled to act as the basis for TESS target selection.¹¹ Because the TIC contains over 1.7 billion point sources globally, querying it requires aggressive spatial and photometric filtering.¹⁴

The system will leverage the specialized subset known as the Candidate Target List (CTL), explicitly designed to identify the optimal stars for exoplanet detection.¹⁵ The Exoplanet CTL version 8.01 (available as a 497 MB CSV file) comprises approximately 9.5 million prioritized

targets, primarily focusing on dwarf stars with optimal TESS-band magnitudes (m_{TESS})

ranging from 3.0 to 17.0, and effective stellar temperatures (T_{eff}) between 2285 K and 10,050 K.¹³ The hackathon mandate suggests downloading a specific sector's high-cadence data, which will yield approximately 20,000 to 30,000 continuous light curves of different stellar targets—a statistically significant sample size for localized model inference and validation.

Photometric Time-Series Acquisition Protocols

TESS observes the celestial sphere in predefined sectors, monitoring each sector continuously for approximately 27-day intervals.¹⁸ The spacecraft operates utilizing multiple observational cadences. While Full Frame Images (FFIs) are acquired at 30-minute, 10-minute, or 200-second cadences depending on the mission cycle, the primary focus for this pipeline will be the highly detailed Target Pixel Files (TPFs) and Light Curve Files (LCFs) extracted at the rapid 2-minute and ultra-rapid 20-second cadences.¹¹

The pipeline will utilize the Python library `lightcurve`, an industry-standard astronomical tool, to programmatically query the MAST servers via the Application Programming Interface (API).¹⁹

The explicit feature targeted for extraction is the Pre-search Data Conditioning Simple Aperture Photometry (PDCSAP) flux.²¹ Unlike standard Simple Aperture Photometry (SAP)—which merely sums the electron counts within a pre-defined pixel mask and leaves the signal heavily contaminated by spacecraft systematics—the PDCSAP flux has been systematically processed by the Science Processing Operations Center (SPOC) pipeline utilizing Cotrending Basis Vectors (CBVs) to effectively mitigate common-mode instrumental anomalies.¹⁸

Auxiliary and Curated Datasets for Model Generalization

While TESS provides the primary testbed, training an AI solely on raw TESS data without a massive catalog of verified ground truths will result in immediate overfitting and failure to meet the evaluation criteria for accuracy. The problem statement notes that a curated dataset of known exoplanets and false positives will be provided, but relying exclusively on this limited provision is a strategic vulnerability. The pipeline will proactively integrate the following auxiliary datasets to enrich the training matrix:

Auxiliary Dataset Source	Data Characteristics and Strategic Utility
NASA Exoplanet Archive	The definitive repository of all confirmed exoplanets. The pipeline will query this database via the Astroquery API to map confirmed TESS Objects of Interest (TOIs) against the raw light curves, establishing an absolute, unquestionable ground truth for the "Genuine Transit" class.
Kepler and K2 Mission Archives	While TESS monitors bright, nearby stars for 27 days, the legacy Kepler mission monitored a single patch of sky continuously for four years. Ingesting Kepler data allows the AI to learn the morphology of multi-planet systems and ultra-long period transits that TESS struggles to capture, serving as a powerful pre-training corpus for transfer learning.
ExoFOP-TESS (Exoplanet Follow-up Observing Program)	A database containing community-driven vetting metrics. This repository is critical for sourcing validated "Eclipsing Binary" and "Stellar Blend" targets. By scraping ExoFOP, the pipeline can construct a massive negative-class dataset, forcing the AI to learn the subtle differences between true planets and high-fidelity impostors.
Synthetic Light Curve Injectors	Utilizing tools like batman or PyTransit, the data pipeline will computationally generate millions of synthetic planetary transits across a wide parameter space (varying depths, durations, and signal-to-noise ratios). ²² These synthetic signals are mathematically injected into raw, noisy TESS light curves that are known to be empty, augmenting the minority class and heavily fortifying the model against extreme class imbalance.

4. High-Performance Preprocessing and Noise Mitigation

Raw PDCSAP light curves, despite undergoing initial SPOC pipeline corrections, remain entirely unsuitable for direct ingestion into a multi-class deep neural network. The presence of residual low-frequency stellar variability, cosmic ray strikes, and long-term momentum dump artifacts will hopelessly confuse the gradient descent process. Therefore, an aggressive, mathematically rigorous preprocessing pipeline must be executed sequentially. Given the massive data volume (tens of thousands of high-cadence light curves), this pipeline must be explicitly parallelized to fully saturate the 24 cores and 32 threads of the latest-generation Intel Core i9 processor.⁵

Step 1: Outlier Rejection and Normalization

The initial algorithmic step involves the aggressive removal of non-physical anomalies and severe stellar flares that cause sudden, localized spikes in the photometric flux. A rolling median filter combined with a robust sigma-clipping algorithm (typically set to an asymmetric threshold of $+3\sigma$ and -7σ) will eliminate high-frequency positive outliers while carefully preserving the negative, downward-facing transit signals.²⁰ Subsequent to outlier rejection, the flux values are normalized such that the median out-of-transit flux equals zero.

$$f_{norm}(t) = \frac{f(t) - \tilde{f}}{\tilde{f}}$$

Where $f(t)$ is the raw flux at time t , and $\tilde{f}$ is the median flux of the light curve segment. This transformation ensures the neural network focuses purely on relative fractional changes in luminosity rather than absolute photon counts, which vary wildly by orders of magnitude across different stellar classifications.²⁴

Step 2: Advanced Baseline Detrending

Low-frequency stellar variability, such as rotational modulation caused by massive starspots, creates quasi-sinusoidal wave patterns in the baseline flux that completely obscure shallow transit events.²⁴ To isolate the high-frequency transit dips, the underlying long-term variability must be mathematically estimated and subtracted.

The pipeline will utilize a generalized non-linear least squares fit implementing basis splines (B-splines) with a carefully tuned break tolerance, or alternatively, a highly aggressive Savitzky-Golay filter. The critical optimization objective here is to flatten the out-of-transit baseline to absolute zero while meticulously ensuring that the high-frequency transit dips are not inadvertently fitted and removed. To achieve this, the detrending algorithm utilizes iterative masking: it identifies potential transit dips, masks them out of the array, fits the spline to the remaining "clean" baseline, and then divides the entire original light curve by the fitted spline.

Step 3: Global Transit Search and Phase Folding

The core feature extraction relies on identifying the underlying orbital period (P) and the transit epoch (T_0 , the exact time of the first transit). This is mathematically executed utilizing the Box-fitting Least Squares (BLS) periodogram, an algorithm explicitly designed to model transits as periodic box-shaped depressions in time-series data.²⁰

$$\text{BLS}(P, T_0, d) = \sum_i w_i (f_i - \text{box}(t_i; P, T_0, d))^2$$

The BLS algorithm searches through a predefined grid of trial periods and durations (d), seeking the combination that minimizes the squared residuals. This is the most computationally expensive CPU operation in the pipeline and must be distributed across the Intel i9's performance cores (P-cores) using vectorized NumPy arrays and the multiprocessing library. Once the optimal period is identified, the entire time-series is phase-folded. Phase folding is a modulo operation that collapses multiple transits occurring over weeks or months into a single, highly pronounced composite transit signal.

$$\phi(t) = \left(\frac{t - T_0}{P} \right) \mod 1$$

This operation vastly improves the Signal-to-Noise Ratio (S/N), aligning the transits directly on top of one another and revealing the true morphological shape of the event, which is critical for distinguishing planets from background blends.¹⁶

Step 4: Multi-View Tensor Generation

To prepare the data for the deep learning architecture, the phase-folded light curve is divided into two distinct tensors, drawing inspiration from proven methodologies in astronomical machine learning.²⁵

1. **The Global View Tensor:** Captures the entire orbital phase (from phase 0 to 1). This is critical for the AI to identify secondary eclipses (a primary indicator of Eclipsing Binaries) and out-of-transit baseline anomalies.
2. **The Local View Tensor:** Provides a highly magnified, high-resolution zoom strictly centered on the primary transit event (e.g., from phase -0.1 to +0.1). This high-fidelity tensor allows the convolutional filters to analyze the exact slope of the ingress/egress boundaries and the curvature of the transit bottom (differentiating the U-shape of a planet from the V-shape of a grazing binary).¹⁰

5. Architectural Framework for Multi-Class

Categorization

The fundamental requirement of Problem Statement 7 is the categorization of light curve dips into four specific classes: Transits, Eclipses, Blends, and Other Astrophysical Categories. To achieve this state-of-the-art multi-class precision, a hybrid deep learning architecture synthesizing 1D Convolutional Neural Networks (CNNs), sequence modeling mechanisms, and gradient boosting ensembles is required.⁸

The Deep Convolutional-Attention Hybrid Network

The core architecture operates as a multi-branch neural network designed to simultaneously process the macroscopic features, the microscopic features, and the tabular stellar metadata.

1. **Global Convolutional Branch:** Ingests the 1D Global View tensor. It utilizes large-kernel 1D convolutions (e.g., kernel size 15, stride 2) followed by Batch Normalization, Rectified Linear Unit (ReLU) activations, and MaxPooling layers. This branch acts as a wide-angle feature extractor, learning to identify alternating odd-even depth differences and secondary dips characteristic of the "Eclipses" class.
2. **Local Convolutional Branch:** Ingests the 1D Local View tensor. It utilizes small-kernel 1D convolutions (e.g., kernel size 3 or 5, stride 1) to extract highly granular morphological features. This branch is strictly optimized to distinguish the sharp V-shapes of "Blends" and grazing binaries from the flat-bottomed U-shapes of genuine "Transits".
3. **Stellar Parameter Branch:** A dense feed-forward network that ingests auxiliary tabular data downloaded from the TIC (e.g., Effective Temperature T_{eff} , Stellar Radius $R_{\star}$, Surface Gravity $\log g$, and Gaia DR3 astrometric excess noise). Supplying the model with the host star's physical parameters provides critical astrophysical context. For example, a 2% deep transit around a massive A-type star implies an object physically larger than Jupiter, highly indicative of an eclipsing binary rather than a planet.

The outputs from these three branches are flattened and concatenated into a unified latent space vector. To capture long-range temporal dependencies and complex phase relationships, this concatenated vector is passed through a multi-head self-attention mechanism, allowing the network to dynamically assign higher statistical weights to the most discriminative features.²⁷

Gradient Boosting Ensemble for Final Categorization

While deep neural networks are unparalleled at latent feature extraction from time-series data, they can sometimes struggle with absolute probabilistic calibration and tabular data integration. To maximize accuracy and adhere to the evaluation criteria, the architecture will function as an ensemble.

The high-dimensional feature embeddings extracted from the penultimate fully connected layer of the CNN/Attention network will be routed away from a standard Softmax classifier and instead fed directly into advanced gradient boosting frameworks—specifically Light Gradient-Boosting Machine (LightGBM) and Extreme Gradient Boosting (XGBoost).²⁸ These

classical, tree-based models excel at constructing complex, non-linear decision boundaries in high-dimensional space and provide highly calibrated probability outputs.²⁶ The final output of the XGBoost classifier yields a probability distribution across the four mandated categories (Transit, Eclipse, Blend, Other), from which the highest probability determines the classification, while the raw probability value itself serves as the required "Confidence Level."

Loss Functions and Class Imbalance Mitigation

The dataset of transiting exoplanets exhibits an extreme, exponential class imbalance. Within any given sector of TESS data, confirmed exoplanets represent less than 0.1% of the total target list, while noise, starspots, and eclipsing binaries dominate the dataset.¹² Training a network on this distribution using standard Cross-Entropy Loss will result in a model that trivially predicts "Other" for every input, achieving 99% accuracy while entirely failing the primary objective.

To systematically force the network to learn the minority class, the pipeline will utilize Focal Loss. Focal Loss dynamically scales the gradient updates based on prediction confidence, heavily penalizing the model for misclassifying the hard-to-detect minority class (Transits) while reducing the gradient impact of the easily classified majority classes.

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

Where p_t is the model's estimated probability for the target class, α_t is a specific weighting factor for class imbalance, and γ is the focusing parameter (typically set to 2.0).

6. Parameter Estimation and Uncertainty Quantification

Detecting the signal and classifying it as a planet is only the first phase. The problem statement explicitly requires the estimation of associated parameters—orbital period, transit duration, and transit depth—along with rigorous significance levels. This quantitative analysis forms the foundation of the mandated 3-page scientific report.

While the neural network handles the categorization, parameter estimation must be grounded in pure statistical physics. The initial estimates derived from the BLS periodogram serve as the starting point, but they lack rigorous uncertainty bounds. To fulfill the hackathon requirements for precision, the pipeline will implement Bayesian Inference via Markov Chain Monte Carlo (MCMC) sampling.

Bayesian Transit Modeling via MCMC

Once the AI flags a light curve as a high-confidence "Transit," the pipeline invokes the exoplanet or batman Python libraries to construct a physical, limb-darkened transit model.¹⁵

The MCMC algorithm maps the posterior probability distribution of the transit parameters given the observed light curve data. The Bayes' theorem formulation is:

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Where θ represents the vector of physical parameters (Period, Epoch, Planet-to-Star Radius Ratio, Impact Parameter, and Limb Darkening Coefficients), D represents the detrended light curve data, $P(D|\theta)$ is the Gaussian likelihood function, and $P(\theta)$ represents the prior probabilities (e.g., utilizing the stellar parameters from the TIC as Gaussian priors). The MCMC sampler (such as the No-U-Turn Sampler, NUTS) explores this multi-dimensional parameter space, taking tens of thousands of steps to find the optimal fit.

Derivation of Required Parameters

From the MCMC posterior chains, the system extracts the following mandated parameters:

- **Orbital Period (P):** Extracted directly from the posterior distribution. The median of the chain serves as the final estimate, while the 16th and 84th percentiles provide the strict 1σ upper and lower uncertainty bounds required for the scientific report.
- **Transit Depth (δ):** Calculated from the squared posterior distribution of the planet-to-star radius ratio $(R_p/R_\star)^2$.
- **Transit Duration (T_{14}):** Analytically derived utilizing the posterior values for the period, the scaled semi-major axis ($a/R_\star$), and the impact parameter (b), solving the geometric duration equation.
- **Signal-to-Noise Ratio (SNR):** The pipeline calculates the formal SNR defined as the transit depth divided by the standard deviation of the out-of-transit flux, multiplied by the square root of the number of data points within the transit duration. An SNR ≥ 10.0 is generally required for a statistically significant detection.¹⁶

The quantification of these uncertainties is not merely an academic exercise; it is an explicit evaluation criterion. Documenting exactly how the MCMC chains converged and how the 1σ confidence intervals were derived will form the core narrative of the submitted 3-page methodology report.

7. Computational Optimization on Intel i9 and NVIDIA RTX 4080 Architecture

The formidable processing requirements of managing 30,000 light curves, training a deep ensemble network, and running Bayesian MCMC samplers must be executed entirely within the

hardware constraints of a local workstation featuring a latest-generation Intel Core i9 processor and an NVIDIA RTX 4080 GPU.⁴ The 30-hour limitation of the Grand Finale necessitates extreme computational optimization to ensure the pipeline can complete multiple training iterations and hyperparameter tuning sweeps.⁶

Maximizing the Intel Core i9 Architecture

Data ingestion, BLS periodogram computation, and spline detrending are fundamentally CPU-bound tasks. The Intel Core i9 (e.g., the 13900K or 14900K series) boasts up to 24 cores and 32 threads, partitioned into Performance cores (P-cores) and Efficient cores (E-cores).⁵ To prevent the CPU from bottlenecking the GPU, the data pipeline will utilize PyTorch's DataLoader with the num_workers parameter heavily optimized to match the available thread count. The preprocessing script will implement the multiprocessing library to distribute the computationally intensive BLS searches across the P-cores. The detrended, phase-folded tensors will be written to an optimized HDF5 file format, heavily compressed in memory, before being streamed across the PCIe bus to the GPU.

Exploiting the NVIDIA RTX 4080 CUDA Capabilities

The NVIDIA RTX 4080 provides 16 GB of ultra-fast GDDR6X VRAM and nearly 10,000 CUDA cores, representing a massive engine for deep learning acceleration.⁴ To maximize training throughput and dominate the 30-hour timeline, the architecture will exploit several advanced hardware optimization techniques.

Hardware Optimization Protocol	Technical Implementation	Impact on Pipeline Efficiency
Automatic Mixed Precision (AMP)	Implementation of PyTorch's torch.cuda.amp.autocast(), dynamically converting 32-bit floating-point (FP32) weights to 16-bit (FP16) during the forward and backward passes. ⁵	Slashes VRAM utilization by nearly 50%, enabling massive batch sizes. Utilizes the RTX 4080's 4th Generation Tensor Cores to accelerate matrix multiplication by up to 300%.
Pinned Memory Transfers	Activating pin_memory=True within the PyTorch DataLoader to lock the host CPU memory.	Bypasses the standard pageable memory, allowing for direct, ultra-high-speed Direct Memory Access (DMA) transfers across the PCIe bus to the VRAM, ensuring the CUDA cores

		are never starved for data.
Gradient Accumulation	Decoupling the physical batch size from the mathematical batch size by accumulating gradients over multiple smaller forward passes before executing the <code>optimizer.step()</code> .	Allows the network to simulate the mathematical stability of processing 2048 light curves simultaneously without physically exceeding the strict 16 GB VRAM limit of the RTX 4080.
CUDA Graph Capture	Tracing the computational graph of the CNN forward pass into a static CUDA execution graph.	Eliminates the microsecond CPU-bound overheads associated with launching individual CUDA kernels, reducing inference time on individual light curves to fractions of a millisecond. ⁴

8. Visualization, Confidence Estimation, and Reporting Protocols

The final evaluation criteria heavily weigh the clarity, accuracy, and visualization of the developed pipeline. The outputs of the AI must be interpretable by human astrophysicists, transitioning the system from a "black box" into an explainable scientific tool.

Visualization Framework

The pipeline will utilize matplotlib and seaborn to generate highly standardized, publication-ready visual diagnostics for every positively identified signal.

1. **Full-time Series Plot:** Displays the raw, 27-day PDCSAP light curve overlaid with the B-spline baseline fit, highlighting exactly where the algorithms detected transit events.
2. **Phase-Folded Diagnostic:** The critical visualization. Displays the phase-folded data zoomed in on the transit epoch. The raw data points are plotted as scatter markers, overlaid with a solid line representing the optimal MCMC physical transit model.
3. **Residual Analysis:** A subplot beneath the phase-folded curve displaying the residuals (Data minus Model). A flat residual line confirms a perfect model fit, demonstrating the mathematical accuracy of the estimated parameters.
4. **Network Activation Mapping:** Utilizing Gradient-weighted Class Activation Mapping (Grad-CAM), the pipeline will generate a heatmap over the 1D light curve, explicitly highlighting which specific data points (e.g., the V-shaped bottom) caused the CNN to classify the object as a 'Blend' rather than a 'Transit'. This directly addresses the need for methodological clarity.

The 3-Page Scientific Methodology Report

The hackathon explicitly requires the submission of a maximum 3-page report detailing the methodology, assumptions, tools used, and uncertainty estimations. This document will be drafted utilizing LaTeX to ensure pristine formatting of mathematical formulas.

The structure of the report will be rigidly defined:

- **Section 1: Data Acquisition & Preprocessing:** Detailing the use of lightkurve, the TESS TIC/CTL, and the mathematics of the B-spline detrending and BLS periodogram.
- **Section 2: AI Architecture & Training:** Explaining the 1D-CNN/Attention hybrid, the XGBoost ensemble, and the use of Focal Loss to overcome the extreme class imbalance between planets and false positives.
- **Section 3: Parameter Estimation & Uncertainty Calculus:** Documenting the Bayesian MCMC sampling framework, explicitly detailing the priors used, the convergence metrics

of the chains, and how the 1σ confidence levels for Period, Depth, and Duration were extracted from the posteriors.

9. Strategic Execution Timeline for the 30-Hour Grand Finale

Given the rigid 30-hour timeframe imposed during the Grand Finale, human capital and computational cycles must be ruthlessly synchronized.⁶ The hackathon rules strictly limit teams to 3-4 undergraduate or postgraduate students.² The team must divide into specialized, parallel operational roles: Data Engineer (Pipeline/Hardware), Machine Learning Architect (PyTorch/CUDA), Domain Analyst (Astrophysics/MCMC), and Deployment Specialist (Visualization/Reporting).

Time Block	Operational Phase	Key Deliverables and Hardware Utilization
Hours 0-5	Infrastructure Initialization and Mass Data Ingestion	The Deployment Specialist configures the RTX 4080 CUDA environments. The Data Engineer triggers the highly parallelized API scripts to download 30,000 light curves from the MAST servers, immediately streaming them into the i9 CPU preprocessing queue.
Hours 5-12	Parallel Preprocessing and Tensor Construction	The Intel i9 executes the B-spline detrending and

		BLS periodogram searches. The Domain Analyst verifies the integrity of the phase-folded outputs. Data is formatted into HDF5 tensors and pushed to the NVMe storage array for rapid GPU access.
Hours 12-20	Network Synthesis and Primary Training Loop	The ML Architect initializes the CNN/Attention architecture. The RTX 4080 executes extreme FP16 mixed-precision training. The cyclical learning rate pushes the model to converge rapidly. Focal Loss mitigates the class imbalance.
Hours 20-25	Ensemble Integration and Parameter Extraction	Deep learning embeddings are routed into the XGBoost classifier. The Domain Analyst triggers the MCMC Bayesian samplers on the positive hits to extract the highly accurate orbital periods, depths, and durations, along with their 1σ confidence limits.
Hours 25-30	Visualization, Report Drafting, and Code Freeze	No further model training occurs. The Deployment Specialist generates the phase-folded and Grad-CAM visualizations. The team collaboratively finalizes the 3-page LaTeX methodology report. The final pipeline is

		containerized and submitted for ISRO evaluation.
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The fusion of deep domain-specific astrophysical preprocessing, advanced dual-branch neural network topologies, rigorous Bayesian uncertainty quantification, and extreme hardware exploitation guarantees a solution characterized by maximum computational efficiency and unassailable scientific validity. Execution of this exhaustive strategic plan will directly fulfill the rigorous demands of Problem Statement 7, establishing a state-of-the-art framework for autonomous astronomical research and securing a dominant position within the Bharatiya Antariksh Hackathon 2026.

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